

Forman Christian College, Lahore

(A Chartered University)

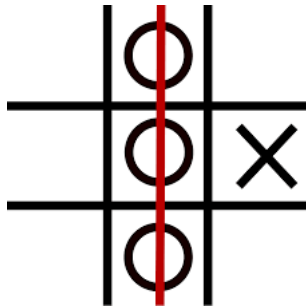
Assignment 1

Objective

To give students practice in using lists and loops

TIC-TAC-TOE

Your little brother loves to play tic-tac-toe (zero kaata) with you, so you decide to build that.



You want him to be able to compete against the computer as well as you. Hence, your game is going to have two modes: (i) single player mode (against the computer) and (ii) two player mode.

In two player mode, your program should first ask the players their names and toss a coin to decide who goes first. Then the program should ask the users to take turns ensure that each move is legitimate. And if any player has won or not.

In one player mode, your program should first ask for the mode. There can be 3 modes: easy, medium, or hard. Then it should toss a coin to decide who gets to go first.

In easy mode, the computer just randomly decides a row and column (really dumb). In hard mode, your program should be good enough so that the human never wins; instead, the game should end in a "draw" or "the computer wins." Here is one possible strategy for selecting the next move:

1. Check if there is a winning move available, and if yes, make that move and win.
2. If not, check if there is a winning move available for the *opponent*, and if yes, take that spot on the board.
3. If not, then check to see if the center spot is empty. If yes, take that spot on the board.

4. If not, then see if there are any open corners on the board. If yes, take the first open corner found on the board.
5. If not, then find any remaining empty spot on the board and take that spot.

Medium should be in between these two. It should not be dumb but not as good as the hard level.

There should be an option to display the last match.

Bonus: When playing against the computer, there should be an option to undo your last move unless the computer has won.

Hint: These last two options can be achieved by storing all the board positions in a list.

Lastly, the menu should repeat unless the user selects the quit option.

1. Single player
2. Two player
3. Display last match
4. Quit

Restrictions

Name the file you create and turn in `TicTacToe.py`. Your program should include a header comment with the following information: your name, course number, section number, assignment title, and date. You should also include comments throughout your code, when appropriate.

Sample Output

1. Single player

2. Two player

D. Display last match

Q. Quit

What do you want to do: **D**

No match found.

1. Single player

2. Two player

D. Display last match

Q. Quit

What do you want to do: **2**

Enter player 1 name: **Nazim**

Enter player 2 name: **Obed**

Tossing a coin to decide who goes first!!!

Obed gets to go first.

```
  |  |  
-----
```

```
  |  |  
-----
```

```
  |  |  
Obed's turn:
```

Obed enter row:**0**

Obed enter col:**0**

```
X |  |
```

| |

| |

Nazim's turn:

Nazim enter row:0

Nazim enter col:0

Invalid move. Nazim try again:

Nazim enter row:1

Nazim enter col:1

X | |

| O |

| |

Obed's turn:

Obed enter row:2

Obed enter col:2

X | |

| O |

| | X

Nazim's turn:

Nazim enter row:1

Nazim enter col:0

X | |

O | O |

| | X

Obed's turn:

Obed enter row:**1**

Obed enter col:**2**

X | |

O | O | X

| | X

Nazim's turn:

Nazim enter row:**0**

Nazim enter col:**2**

X | | O

O | O | X

| | X

Obed's turn:

Obed enter row:**2**

Obed enter col:**0**

X | | O

O | O | X

X | | X

Nazim's turn:

Nazim enter row:**2**

Nazim enter col:**1**

X | | O

O | O | X

X | O | X

Obed's turn:

Obed enter row:**0**

Obed enter col:**1**

X | X | O

O | O | X

X | O | X

It's a draw!

1. Single player

2. Two player

D. Display last match

Q. Quit

What do you want to do: **D**

Obed (X) vs Nazim (O)

| |

| |

| |

Obed:

X | |

| |

| |

Nazim:

X | |

| O |

| |

Obed:

X | |

| O |

| | X

Nazim:

X | |

O | O |

| | X

Obed:

X | |

O | O | X

| | X

Nazim:

X | | O

O | O | X

| | X

Obed:

X | | O

O | O | X

X | | X

Nazim:

X | | O

O | O | X

X | O | X

Obed:

X | X | O

O | O | X

X | O | X

It's a draw!

1. Single player

2. Two player

D. Display last match

Q. Quit

What do you want to do: **1**

E: easy

M: medium

H: hard

Which level do you want to play: **M**

Enter player 2 name: **Obed**

Tossing a coin to decide who goes first!!!

Obed gets to go first.

| |

| |

| |

Obed's turn:

Obed enter row:**0**

Obed enter col:**0**

```
X |  | 
-----
  |  | 
-----
  |  |
```

Computer's turn:

```
X |  | 
-----
  | O | 
-----
  |  |
```

Obed's turn:

Obed enter row:2

Obed enter col:2

```
X |  | 
-----
  | O | 
-----
  |  | X
```

Computer's turn:

```
X |  | 
-----
O | O | 
-----
  |  | X
```

Obed's turn:

Obed enter row:**1**

Obed enter col:**2**

X			
---	--	--	--

O		O		X
---	--	---	--	---

				X
--	--	--	--	---

Computer's turn:

X				O
---	--	--	--	---

O		O		X
---	--	---	--	---

				X
--	--	--	--	---

Obed's turn:

Obed enter row:**2**

Obed enter col:**0**

X				O
---	--	--	--	---

O		O		X
---	--	---	--	---

X				X
---	--	--	--	---

Computer's turn:

X				O
---	--	--	--	---

O		O		X
---	--	---	--	---

X		O		X
---	--	---	--	---

Obed's turn:

Obed enter row: **0**

Obed enter col: **1**

X | X | O

O | O | X

X | O | X

It's a draw!

1. Single player

2. Two player

D. Display last match

Q. Quit

What do you want to do: **Q**

Thanks for playing. Hope you had fun!

Submission instructions

You need to submit the file, TicTacToe.py.

Notes on Grading

- All variables should have informative names.
- Your input/output should match the examples.
- Every program must have comments. There should be header comments in your program including your name and a brief introduction of the program. You should have at least one comment besides that.
- Any plagiarism/cheating would be strictly dealt with.
- 1 day late assignment would be penalized by a 10% grade deduction. 2 days late would be penalized by 25% grade deduction. Assignments submitted 2 days after the submission date would not be accepted.
- Failure to send TicTacToe.py would result in a zero for the assignment.